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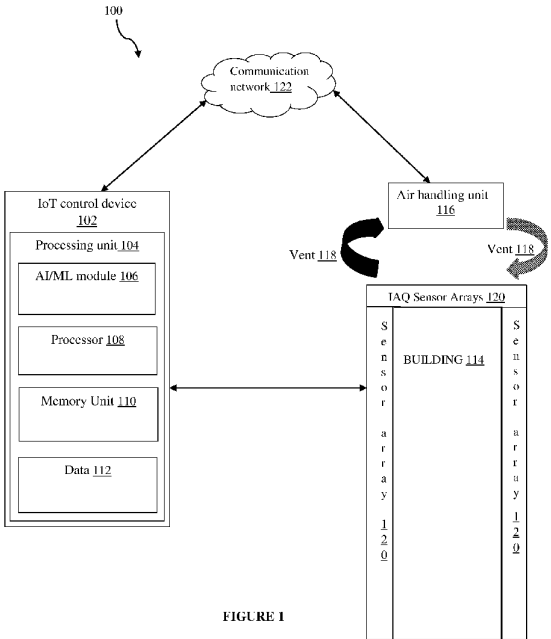
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(54) **Title:** A SYSTEM AND METHOD FOR AIR QUALITY MANAGEMENT



(57) **Abstract:** A system (100, 200) for air purification in a building, the system comprising a plurality of air quality sensor arrays (120) installed in a plurality of zones in the building (114) to provide air quality data from one or more zones in the building, an IoT-managed control device (102) configured to receive sensor data from said plurality of air quality sensor arrays (120) and process the same to provide one or more command signals, wherein the IoT-managed control device (102) processes the air quality sensor data from each zone in the building (114) and generates a command signal for the supply of a corresponding amount of purified air at a desired temperature to each of the zones, an air handling unit (116) configured to receive the one or more command signals from the IoT-managed control device (102) regarding the quality of air in the one or more zones, and provide purified air to said one or more zones through air circulation lines, wherein the air handling unit (116) receives vent air (118) at a first temperature as input and provides purified air at a second temperature, wherein the IoT-managed control device (102) comprises an AI/ML module (106) configured to predict air quality within the one or more zones of the building based on the processing of the sensor data, at least one piece of forecast data received from one or more sources over a network (122), and historical data, and optimizes the energy consumption within the building (114) based on predicted data.

TITLE OF THE INVENTION**“A SYSTEM AND METHOD FOR AIR QUALITY
MANAGEMENT”**

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FIELD OF THE INVENTION

[0001] The present invention generally relates to air conditioning and purification systems and in particular relates to an air quality management system and method for spaces such as apartments and buildings, schools, malls, hospitals, residential and commercial buildings.

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BACKGROUND OF THE INVENTION

[0002] Indoor air quality can be three to five times more polluted than outdoor air quality. In the Middle East, specifically, indoor air quality is poor. The quality of indoor air directly impacts the health, wellness, and cognitive ability of residents. Sick Building Syndrome (SBS), which refers to a condition believed to be triggered by poor indoor air quality, is a common phenomenon among residents in many countries nowadays.

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[0003] While the exact cause of SBS may be challenging to pinpoint, potential factors include poor ventilation, high dust levels, tobacco smoke, mold or fungus presence, chemicals from cleaning products, pesticides, high-stress levels, and low humidity in enclosed spaces.

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[0004] The significance of indoor environmental quality, encompassing factors like visual, acoustical, thermal comfort, and indoor air quality, significantly impacts health, productivity, and performance, particularly in educational and office settings. Overall, addressing indoor air quality not only enhances health and comfort but also contributes to energy efficiency. The implementation of innovative solutions can reduce energy consumption, resolve SBS, and create a healthier indoor environment for occupants.

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[0005] US patent publication 11,635,221 B2 discloses an air-purification, remote HVAC management, and indoor air quality monitoring system including an online cloud-based platform. The system may be a locally based system that utilizes a specially designed Artificial Intelligence

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platform for optimizing energy efficiency, enthalpy, and air quality based on continual sensor data collection, indoor air quality measurements, and dynamically adjusted operating parameters. A plurality of sensor units are configured to detect and measure CO, CO₂, Sulfur Dioxide, Nitrogen Dioxide, Ozone, and other gas concentration levels, and can monitor building pressure. A fault indicator and communications display may also be included.

[0006] US patent publication US20220203288 A1 discloses systems and methods of holistically controlling a heating, ventilation, air conditioning, and refrigeration (HVACR) system and a multi-contaminant air cleaner (MCAC). The method includes sensing, via at least one HVACR sensor, at least one comfort parameter; sensing, via at least one MCAC sensor, at least one air-quality parameter; determining, via a controller, minimum air changes of the HVACR system based on the at least one comfort parameter; and determining, via the controller, equivalent air changes of the MCAC and an indoor air quality (IAQ) score based on the at least one air-quality parameter. The method also includes controlling the operation of the HVACR system and controlling the operation of the MCAC based on the determined air changes to optimize energy consumption. The publication also suggests that the sensed, measured, or determined data/parameters can be recorded, stored, and/or used for predicting and deploying the next control configuration/logic using artificial intelligence or machine learning.

[0007] Further, US20220268475 A1 discloses an indoor air quality control system to control a plurality of air handling units within an industrial facility in a concerted effort to effect an overall air quality goal. A remote server analyzes sensor data, historical data, and other environmental data (e.g., predicted weather data), and uses one or more machine learning algorithms to model the behavior of air within the facility. A plurality of sensors deployed within the facility each variously measure environmental values within a respective sensor's range that are relevant to air quality, such as humidity, temperature, air pressure, level of contaminants, and the like. The sensed air quality data is considered holistically to understand the overall condition of the facility and the gradient of air flows and/or

contaminant flows within the 3-dimensional space. Air handling models are applied to current sensor data to generate instructions to selectively turn on/off or otherwise control components of various air handling equipment to reach an optimized air quality result. Decisions on how to control the facility are based on environmental health and safety considerations. In an embodiment, an environment modeling logic may obtain data from one or more third-party servers, which may include a weather prediction or analysis, such as a determination of predicted atmospheric conditions, outside temperature, and pollution forecast.

[0008] Further, in a non-patent document titled, "Indoor Air Pollution, Related Human Diseases, and Recent Trends in the Control and Improvement of Indoor Air Quality", Apr 2020, Tran et al talk about Internet-of-Things (IoT), big data, machine-learning technologies being introduced as trending technologies that offer great capability for real-time IAQ monitoring. One of the important applications of IoT in IAQ monitoring is "electronic noses" (E-noses), which are biomimetic-type devices that mimic the functionalities of mammals' olfaction systems. Generally, an E-nose system consists of four basic components: (i) A multi-sensor array; (ii) software with digital pattern recognition algorithms; (iii) an information-processing unit (i.e., an artificial neural network (ANN)), and (iv) reference library databases. By using the chemical sensor arrays in combination with the classification algorithms, E-noses can easily monitor target gases by detecting and discriminating types and concentrations.

[0009] In addition to the above, the document also talks about smart devices based on the integration of cloud computing and IoT to precisely monitor IAQ and efficiently transmit real-time data to a cloud computing-based web server using an IoT sensor network. The Smart-Air device may include a pollutant detection sensor array, a microcontroller, and an LTE modem. The sensors in the Smart-Air device include a VOC sensor, a laser PM sensor, a CO₂ sensor, a CO sensor, and a temperature/humidity sensor. The Smart-Air platform additionally relies on cloud computing technology and IoT technology to monitor and control IAQ anytime and anywhere.

5 [0010] Conventional air quality handling systems incur significant energy costs owing to cleaning intake air throughout their working period, plus the maintenance cost of cleaning the ducts is also considerable. Lack of cleaning leads to severe mold and fungal growth inside the ducts, again posing a health hazard in the indoor environment.

[0011] Further, conventional systems lack management of air quality across different zones in premises, the use of waste heat and/or cooling equipment already installed, and are limited to predicting air quality based on sensor data.

10 [0012] Conventional fresh air handling units (FAHU) are designed for individual buildings. However, these are not scalable and work as a centralized system that provides clean air for an entire community or city like a district cooling system.

15 [0013] Therefore there is a need to provide a solution for healthy indoor air quality, reducing or eliminating SBS and energy consumption. There is also a need to develop a low-cost, highly scalable solution for substantial reduction of carbon footprint and electricity bills, reduction of heat island impact, and reduction in maintenance costs.

20 [0014]
OBJECT OF THE INVENTION

[0015] An objective of the present invention is to provide a system and method for efficient air purification in different zones of a building.

25 [0016] Another objective of the present invention is to provide an intelligent air purification system that is coupled with a chiller system or a district cooling system for individual villas, villa communities, townships, entire city, hospitals, or buildings.

[0017] An additional objective of the present invention is to provide an intelligent air purification system, removing or reducing Fresh air handling systems capacity, size, and 30% of energy consumption that conserves energy not only in cooling but also, more significantly, in dehumidification, resulting in a very short payback period.

30 [0018] Yet another objective of the present invention is to provide an intelligent air purification system that can predict ambient air quality and operate at optimal levels.

[0019] Another objective of the present invention involves transferring conditioned air through a system designed to resemble pipes and fountains, thereby reducing the substantial costs associated with duct-based air distribution systems and mitigating maintenance issues.

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SUMMARY OF THE INVENTION

[0020] It will be understood that this disclosure is not limited to the particular systems, and methodologies described, as there can be multiple possible embodiments of the present disclosure which are not expressly illustrated in the present disclosure. It is also to be understood that the terminology used in the description is to describe the particular versions or embodiments only, and is not intended to limit the scope of the present disclosure.

[0021] Described herein is a system and method for air purification in buildings. In one embodiment, the system comprises multiple air quality sensor arrays installed within a building to provide comprehensive air quality data. An IoT-managed control device is set up to receive sensor data from multiple air quality sensor arrays and process this information to generate one or more command signals. This device analyzes air quality sensor data from individual zones within the building and produces command signals, instructing the supply of an appropriate volume of purified air at desired temperatures to each zone.

[0022] An air handling unit is configured to receive these command signals from the IoT-managed control device, monitoring the air quality in the respective zones. It supplies purified air to these zones through air circulation lines. The air handling unit takes in vented air at an initial temperature and outputs purified air at a different, regulated temperature.

[0023] Furthermore, the IoT-managed control device includes an AI/ML module designed to forecast air quality requirement within the building's zones. The AI/ML module utilizes sensor data processing and forecast information obtained from various sources via a network. By collaborating with the air handling unit (also referred to as a Smart Air Handling Unit or SAHU), it optimizes energy consumption within the building based on predictive data.

- 5 [0024] Another embodiment of the system reveals that the air quality sensor arrays comprise at least one sensor chosen from each of the following groups: particulate matter sensors, volatile organic compound (VOC) sensors, carbon dioxide (CO₂) sensors, Nitrogen Oxide (NO₂) sensors, Sulphur Oxide (Sox), Ozone, Radon temperature sensors, humidity sensors and air pressure sensors.
- 10 [0025] One embodiment of the present invention discloses that the AI/ML module is designed to dynamically adjust one or more command signals for each zone. This adaptation is based on historical air quality data and predefined standard preferences specific to each zone.
- 15 [0026] Another embodiment discloses that the air handling unit is programmed to modulate the air purification level according to the severity of detected air pollutants in a zone. This adjustment is guided by the received one or more command signals about the respective zone.
- 20 [0027] An embodiment further details that the AI/ML module predicts potential faults or inefficiencies in the air handling unit by analyzing processed sensor data. This proactive approach facilitates preemptive maintenance, ensuring sustained optimal air purification performance.
- [0028] Another embodiment highlights the IoT-managed control device, incorporating a user interface for real-time air quality monitoring, purification settings adjustment, and receipt of notifications or recommendations derived from AI/ML module predictions.
- 25 [0029] Additionally, the system comprises a fresh air handling unit capable of operating in isolated or standby modes alongside the primary air handling unit. This feature allows periodic intake of fresh air into the system.
- 30 [0030] Furthermore, the air handling unit consists of a fan-coil unit adapted for different modes of operation. In one mode, it conditions stale air from multiple zones of the building at varying temperatures using hot or cold water circulation. In another mode, it purifies and conditions air based on sensor data.
- [0031] The system's embodiments reveal the air handling unit's utilization of waste heat produced within the building for water heating or cool vent air to regulate water temperature.

[0032] Moreover, the fan-coil unit is linked with at least one air recycle compressor, a knockout drum to separate moisture from air, and air pressure-reducing valves. These components release purified and processed air at a lower temperature into the building's zones.

5 [0033] The air quality sensor arrays encompass exhaust air quality meters, continuous or clean air quality meters, and indoor air quality meters.

[0034] Additionally, the system integrates an energy meter and a thermal unit meter. The processing unit regulates building power consumption based on energy consumption data.

10 [0035] The air quality sensor arrays are installed both inside and outside the building. The IoT-managed control device triggers an alarm module if readings from an array in one zone significantly vary from corresponding zones beyond a preset threshold.

15 [0036] In another embodiment, the invention details a method for air purification in a building utilizing air quality sensor arrays, an IoT-managed control device, and an air handling unit. The system processes air quality data from installed sensor arrays within and outside the building. The IoT-managed control device analyzes the data to generate one or more command signals for each zone, directing the supply of purified air to
20 individual zones. These signals are transmitted to the air handling unit through which purified air is delivered to multiple zones via circulation lines. The AI/ML module predicts air quality within zones using sensor and forecast data obtained from the air handling unit, optimizing energy consumption in the building based on its predictions to enhance air quality
25 within the zones.

[0037] Various objects, features, aspects, and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which the same numerals represent like components.

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BRIEF DESCRIPTION OF THE DRAWINGS

[0038] Non-limiting examples of the present disclosure will be described in the following disclosure with reference to the appended drawings, in which:

[0039] Figure 1 illustrates a block diagram depicting an exemplary architecture of an air purification system, as per an embodiment of the present invention;

5 [0040] Figure 2 illustrates a block diagram demonstrating the implementation of the system of Fig. 1;

[0041] Figure 3 illustrates a block diagram of the air handling unit of the air purification system of Fig. 1, according to an embodiment of the present invention;

10 [0042] FIG. 4 illustrates a block diagram of the system implemented for air purification, either in an individual building or in a cluster of buildings, according to an embodiment of the present invention; and

[0043] FIG. 5 illustrates a flow diagram of a method for air purification in a building using the system, according to an embodiment of the present invention.

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DETAILED DESCRIPTION OF THE DRAWINGS

[0044] Some embodiments of this invention, illustrating all its features, will now be discussed in detail. The words “comprising,” “having,” “containing,” and “including,” and other forms thereof, are intended to be
20 equivalent in meaning and be open-ended in that an item or items following any one of these words is not meant to be an exhaustive listing of such item or items, or meant to be limited to only the listed item or items.

[0045] It must also be noted that as used herein and in the appended
25 claims, the singular forms “a,” “an,” and “the” include plural references unless the context dictates otherwise. Although any systems and methods similar or equivalent to those described herein can be used in the practice or testing of embodiments of the present invention, the preferred, systems and methods are now described.

30 [0046] The present invention discloses a system and method for air purification in buildings. The system incorporates an air handling unit aimed at improving air quality within the building. The air handling unit receives command signals from the IoT-managed control device, which analyzes sensor data, historical records, and information from various

sources. Using multiple machine learning algorithms, it models the behavior of air within the building.

[0047] Additionally, the system can effectively manage air quality by simultaneously regulating the temperature. This approach optimizes energy consumption based on predicted data, applicable in either a single building or a cluster of buildings. It addresses contaminants, pollutants, moisture, temperature variations, and volatile organic compounds.

[0048] The sensed air quality data is comprehensively evaluated to grasp the overall conditions in different zones of the building and to understand the gradients of air and/or contaminant flows within the three-dimensional space. The current system for air handling utilizes real-time sensor data and weather forecasts to generate instructions. These instructions selectively activate or deactivate the air handling unit to achieve optimized air quality. Decisions regarding the control of the building's air quality prioritize environmental health and safety considerations.

[0049] FIG. 1 illustrates a block diagram 100 of the architecture of an air purification system in a building, in accordance with an embodiment described in the disclosure. As shown, the system primarily consists of an IoT-managed control device 102 communicatively coupled to an air handling unit 116 over a network 122 to manage air quality within a building 114. A plurality of air quality sensor arrays 120 is positioned within the building 114 to collect air quality data from various zones within the building, both inside and outside of the building 114, not limited to specific surfaces like walls, ceilings, or floors due to their wireless nature. Each sensor array may include, for example, air quality sensors adapted to measure various gases such as CO₂ and CO, VOC, NO₂, SOX, Ozone, Radon, particulate matter, air pressure, temperature, humidity, and the like as well as electric current sensors to measure energy consumption of air conditioning and other equipment within the given premises. The sensors can be deployed anywhere in a three-dimensional space, including on columns, or may be concealed within surfaces. Electric current sensors, in an embodiment, are installed on various units such as exhaust fans, de-stratification fans, water cooling/heating units, and other air handling or conditioning units to monitor their status. The

electric current sensor's assessment of air quality reflects the condition of one of the multiple air handling units 116, or SAHU 116, if they are working properly. Regular data collection occurs from these sensors, which can vary in number and type based on the building size and requirements. Additionally, not all buildings/facilities may require every sensor type, and the systems outlined here have the flexibility to gather data from a variety of sensors and issue commands to any air handling or conditioning units accordingly.

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[0050] In an embodiment, the IoT-managed control device 102 receives data from these sensor arrays 120 in real-time and processes the same to generate one or more command signals. The control device 102 analyzes air quality sensor data from each zone in the building 114 and generates command signals, directing the supply of an appropriate volume of purified air at a desired temperature to each zone based on the command signal(s).

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[0051] In an embodiment, the air handling unit 116 receives the command signals from the IoT-managed control device 102, indicating the air quality in the respective zones. Subsequently, it supplies purified air to these zones, for example, through air circulation lines (pipes). In an embodiment, the air handling unit 116 takes vent air at an initial temperature as input and provides purified air at a regulated second temperature. In this embodiment, the system keeps the supply of fresh air coming from outside closed and recirculates the vent air, which in summers, for example, may be at a lower temperature than the air outside the building, to various zones of the building after purification. In an embodiment, the present invention involves transferring conditioned air through a system designed to resemble pipes and fountains in place of using the existing duct system within the building. This approach reduces the energy cost substantially as the system utilizes only the relatively cooler vent air and supplies the same through a piping system, thus mitigating maintenance issues related to the ducts. In an embodiment, the ducts may be used intermittently to supply fresh from outside, which supply may again be connected to the air handling unit 116, which then purifies the air and regulates its temperature for use within all or certain

zones of the building 114. In an embodiment, the air handling unit 116 employs the exchange of cold water or the use of waste heat within the building to either lower or increase the temperature of the circulating air.

5 [0052] In an embodiment, the IoT-managed control device 102 incorporates an AI/ML module 106 which runs a prediction algorithm to predict air quality within the building's zones. The AI/ML module 106 takes into account sensor data as well as historical air quality data stored within the system and forecast data received over a network 122 from various sources as input, such as news sources and social media. By
10 collaborating with the air handling unit 116, the device 102 optimizes energy consumption within building 114 based on the predicted data generated by the AI/ML module. In an embodiment, the AI/ML module 106 includes a dedicated Processing Unit involving hardware for AI/ML computations, such as a neural processing unit (not shown) and interfaces
15 to receive data including sensor data. The AI/ML model parameters may be stored and updated through a memory unit 110. The received data is normalized for processing using known data pre-processing techniques.

[0053] In an embodiment, the device 102 integrates AI/ML capabilities for predicting air quality within different zones of the building by analyzing
20 sensor data, historical data, and weather forecast information and optimizes energy consumption by manipulating the control signal governing the air quality management equipment within the building 114, thus ensuring an optimal distribution of fresh air in a building zone according to the requirement. This is required as not all zones in a building
25 would require the same amount of purified air all the time, for example, the reception/entrance area of the building. The system may thus handle equipment such as air conditioning units, air cutters, water chilling machines, vents, and air blowers based on a predictive analysis of the AI/ML module.

30 [0054] Figure 2 illustrates a block diagram showcasing an exemplary system designed for air purification within a building, in accordance with an embodiment of the present invention. System 200 as shown includes a fresh air handling unit 206, which can be operated in isolation or standby

mode alongside the air handling unit 116, enabling the intermittent intake of fresh air into a premises, shown for example, by reference numeral 114.

[0055] The system 200 further includes a fan-coil unit 204 configured to condition vent air received from the premises 114 at a given temperature using cold water circulation received from a water unit 214, and then purify the temperature-conditioned air based on sensor data, for example, as received from an indoor air quality meter 216. In an embodiment, the system 200 may utilize waste heat produced within the building to heat the water of the unit working in off mode, for example, by extracting heat from relatively warmer air circulating within a duct system of the premises 114. Further, optionally, the system 200 may include an air heating system as well.

[0056] In addition, the system comprises an exhaust air quality meter 208 to measure the quality of the vent air and a continuous or clean air quality meter 210 to measure air quality after purification by the air handling unit 116. The system 200 is integrated into the IoT-managed control device 102 and the meters 208, 210, and 216 through data network 202. The system 200 assesses the air quality in various zones within the premises 114 and provides a command signal for the air handling unit 116 to provide clean, conditioned air to said zones.

[0057] As the outside air supply into the premises 114 is blocked and only the vent air, which is cooler in comparison to outside air in summer, is being conditioned and filtered, the energy consumption within the premises 114 is greatly reduced. In winter, the waste heat produced within the building from areas such as kitchens, living rooms, hot water spas, and the like, is captured through the ducts and transferred to heat the vent air being recirculated to all the parts of the premises 114. For such a scenario, suitable heat-trapping and transfer mechanisms may be deployed.

[0058] The system 200 is further connected to an energy meter 218 and a thermal unit meter 212 and is adapted to receive readings from said meters to regulate the power consumption in the building 114 based on the current air quality data. For example, if the vent air is already cool to a level as desired, or purified to a reasonable extent, i.e., within a threshold level of air quality, the system 200 may shut the air handling unit 116 or the water

unit 214 to stop cooling (or heating, as may be the case) the water for a certain period till the air quality drop below the desired quality.

[0059] In an embodiment, the IoT-managed control device 102 is programmed to trigger an alarm module if readings from a sensor array/air quality meter in one zone deviate beyond a preset threshold, or compared to the readings of sensor arrays in similar zones in the premises 114. This may also be an indication of a sensor array or a meter malfunction. The air handling unit 116 is adapted to function along with the fresh air handling unit 206 working in an isolated and standby mode, facilitating occasional intake of fresh air into the system. The air purification system 200 capable of reducing energy consumption by 30% in the isolated/standby mode. This conservation is evident not only in cooling but also in dehumidification, leading to a remarkably short payback period.

[0060] Figure 3 illustrates a diagram representing an exemplary implementation of the air handling unit 116 with the fresh air handling unit 206 in the isolated/standby mode, according to an embodiment. In said mode, the fan-coil unit 204 is coupled to at least one air recycle compressor 314, a knockout drum 308, a purification unit 306, a turbine 304 and one or more air pressure-reducing valves 302..

[0061] Through these, the air handling unit 116 provides purified and processed air at a lower temperature to the various zones of the building 114 while the fresh air handling unit 206 remains isolated or in standby mode, to intake fresh air only when required through an opening of intake vents coming to the fresh air handling unit 206. In the mode, as shown, when using the system 200 in summertime, the air handling unit 116 intakes vent air 118 at a first temperature, compresses it via the compressor 314, and passes the compressed air to a water circulation unit 310, where the vent air comes in indirect contact with cool water, thus lowering the temperature to a certain level. The air from the water circulation unit 310 is sent to the purification unit 306, where all the impurities and pollutants are removed from the air and the purified air is sent back to the premises, via pipes to the premises 114. Moreover, the knockout drum 308, where moisture is extracted from the air received at a certain pressure. The turbine 304 receives 8 bar pressurized air from the

KO Drum 308 that is used to generate electricity. As the rest of the released air is pressurized, one or more pressure-reducing valves 302 are used to lower the pressure, and eventually cooling the air further before being dispatched to the premises 114. Waste air from the purification unit 306 is purged outside of the system to the outside environment.

[0062] FIG. 4 presents a schematic representation of the air purification system, capable of effectively managing and regulating air quality within a single building or a group of interconnected buildings. This exemplary system demonstrates a robust approach to maintaining and controlling air quality parameters for enhanced comfort and health of occupants.

[0063] At its core, the system is anchored by an interconnected communication network that gathers data from various sensors installed in each building. Each building within this network is equipped with its dedicated air handling unit 116. These air handling units 116 are controlled by the IoT-managed control device(s) 102 that serve as the control centers, effectively overseeing and managing the air quality parameters within the respective buildings. The air quality parameters within buildings or occupant areas are regulated through the implementation of a synchronization loop in the air purification system. This synchronization loop operates via various control units designed for the removal of pollutants or the maintenance of humidity levels. The control units include the air handling unit 116, which is managed by the IoT-controlled device 102, and control valves 402 such as pressure-reducing valves, utilized to decrease pressure. Additionally, the fan coil unit responsible for conditioning the vented air received from the occupant areas. This process is replicated across multiple occupant areas to consistently maintain preset air quality parameter values.

[0064] Consequently, the synchronization loop is iterated to effectively regulate and control the recycled air quality within each occupant area.

[0065] The IoT-managed control device 102 collects data from multiple sensors placed within buildings and/or from different weather channels and analyzes the quality of the ambient/indoor air in each building. The sensed/detected parameters (data) can be recorded, stored, and then processed by the AI/ML module 106 integrated into the IoT-managed

control device 102 to predict air quality within several such buildings based on the processing of sensor data, forecast data, and historical data. The IoT-managed control devices 102 ensure uniformity and synchronization in the management of air quality parameters throughout the entire cluster of buildings. By implementing a centralized control mechanism, the system streamlines the monitoring and regulation of air quality, optimizing efficiency and effectiveness.

[0066] The system is designed to address various aspects crucial for maintaining superior air quality in different zones of the building. It incorporates mechanisms for air filtration, ventilation, temperature control, humidity regulation, and potentially other customizable features tailored to specific environmental requirements. The integration of these functionalities within each building's air handling unit 116 ensures a comprehensive approach to creating a healthy and comfortable indoor environment.

[0067] The network of an exemplary system harnesses renewable energy and utilizes captured carbon in a greenhouse cooled by radiant cooling, powering the network sustainably.

[0068] These embodiments serve as the foundation for scaling systems in terms of size, functionality, and complexity and accommodating a wide range of processing options such as processing, sub-processing, and byproduct separation of various natural and artificial organic and inorganic compounds. The system, for example, involves electronic monitoring, identification, energy generation, and the ability to respond to base load energy requirements, thereby ensuring grid stability by compensating for supply deficiencies and meeting end-user demands. These elements, identified above, contribute to the system's distinctive features.

[0069] More specifically, the preferred system in the present invention involves a connected, integrated, and controlled energy system employing adaptable hybrid processes. This aims to promote and sustain sustainable operations while effectively managing energy and environmental controls to maintain an efficient production of high-yield products and byproducts. The system operates using renewable energy sources and has backup power from energy storage elements to ensure continuity.

[0070] Furthermore, the system offers flexibility and scalability, allowing for seamless integration into both individual buildings and larger clusters of interconnected structures. Its adaptability makes it suitable for diverse architectural layouts and building types, offering a versatile solution for ensuring optimal air quality management in varied settings.

[0071] Overall, the depicted system in FIG. 4 exemplifies a sophisticated and comprehensive approach to air quality management, showcasing a networked system that efficiently regulates and controls air quality across multiple buildings, promoting healthier and more comfortable indoor environments for occupants.

[0072] FIG. 5 illustrates a flow diagram depicting a method of air purification in a building, in accordance with an embodiment of the present invention. In operation, an IoT-managed control device 102 gathers data on various factors influencing air quality within the building from one or more sensor arrays 120. This includes acquiring sensor data related to indoor and outdoor environmental conditions, such as temperature, humidity, air pressure, and pollutants such as VOC and CO₂ and the like, at step 502. Additionally, the device 102 can access historical weather/air quality records stored therein and utilize the same along with real-time weather forecast data received from various sources including news channels and social media to give predictions on air quality control through an AI/ML module.

[0073] The received air quality data undergoes processing using the IoT-managed control device 102, at step 504. The processing generates one or more command signals based on the processed air quality data received from each zone in the building. These signals include information on the supply of an appropriate amount of purified air to each zone. Subsequently, the generated command signals are transmitted from the IoT-managed control device to an air-handling unit, at step 506.

[0074] The air handling unit or SAHU 116 receives the command signals regarding the air quality in multiple zones and provides purified air to these zones, for example, through dedicated air circulation lines, at step 508.

[0075] The AI/ML module predicts air quality within the multiple zones of the building by processing sensor data, historical weather/air quality data stored in the system, and forecast data obtained from the air handling unit to provide predictions, at step 510, which is used to optimize energy consumption within the building through various energy devices and equipment running in the building. The optimization strategy enhances air quality in different zones based on the specific air quality requirement in each of said zones.

[0076] Embodiments of the present invention may be provided as a computer program product, which may include a computer-readable medium tangibly embodying thereon instructions, which may be used to program a computer (or other electronic devices) to perform a process. The computer-readable medium may include but is not limited to, fixed (hard) drives, magnetic tape, floppy diskettes, optical disks, compact disc read-only memories (CD-ROMs), magneto-optical disks, semiconductor memories, such as ROMs, random access memories (RAMs), programmable read-only memories (PROMs), erasable PROMs (EPROMs), electrically erasable PROMs (EEPROMs), flash memory, magnetic or optical cards, or other type of media/machine-readable medium suitable for storing electronic instructions (e.g., computer programming code, such as software or firmware). Moreover, embodiments of the present invention may also be downloaded as one or more computer program products, wherein the program may be transferred from a remote computer to a requesting computer by way of data signals embodied in a carrier wave or other propagation medium via a communication link (e.g., a modem or network connection).

[0077] In various embodiments, the article(s) of manufacture (e.g., the computer program products) containing the computer programming code may be used by executing the code directly from the computer-readable medium or by copying the code from the computer-readable medium into another computer-readable medium (e.g., a hard disk, RAM, etc.) or by transmitting the code on a network for remote execution. Various methods described herein may be practiced by combining one or more computer-readable media containing the code according to the present invention with

appropriate standard computer hardware to execute the code contained therein. An apparatus for practicing various embodiments of the present invention may involve one or more computers (or one or more processors within a single computer, or one or more processor cores) and storage systems containing or having network access to computer program(s) coded following various methods described herein, and the method steps of the invention could be accomplished by modules, routines, subroutines, or subparts of a computer program product.

[0078] While for purposes of simplicity of explanation, the illustrated methodologies are shown and described as a series of blocks/steps, it is to be appreciated that the methodologies are not limited by the order of the blocks, as some blocks can occur in different orders and/or concurrently with other blocks from that shown and described. Moreover, less than all the illustrated blocks may be required to implement an example methodology. Blocks may be combined or separated into multiple components. Furthermore, additional and/or alternative methodologies can employ additional, not illustrated blocks.

[0079] In the foregoing description, certain terms have been used for brevity, clearness, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed. Therefore, the invention is not limited to the specific details, the representative embodiments, and the illustrative examples shown and described. Thus, this application is intended to embrace alterations, modifications, and variations that fall within the scope of the appended claims.

[0080] The methodology and techniques described for the exemplary embodiments can be performed using a machine or other computing device within which a set of instructions when executed may cause the machine to perform any one or more of the methodologies discussed above. In some embodiments, the machine operates as a standalone device. In some embodiments, the machine may be connected (e.g., using a network) to other machines. In a networked deployment, the machine may operate in the capacity of a server or a client-user machine in a

server-client-user network environment, or as a peer machine in a peer-to-peer (or distributed) network environment.

[0081] Moreover, although the present invention and its advantages have been described in detail, it should be understood that various changes, substitutions, and alterations can be made herein without departing from the invention as defined by the appended claims. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, means, methods, and steps described in the specification. As one will readily appreciate from the disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

[0082] The preceding description has been presented with reference to various embodiments. Persons skilled in the art and technology to which this application pertains will appreciate that alterations and changes in the described structures and methods of operation can be practiced without meaningfully departing from the principle, spirit, and scope.

CLAIMS**We claim:**

1. A system (100, 200) for air purification in a building, the system comprising:
 - a plurality of air quality sensor arrays (120) installed in a plurality of zones
 - 5 in the building (114) to provide air quality data from one or more zones in the building;
 - an IoT-managed control device (102) configured to receive sensor data from said plurality of air quality sensor arrays (120) and process the same to provide one or more command signals, wherein the IoT-managed control device
 - 10 (102) processes the air quality sensor data from each zone in the building (114) and generates a command signal for the supply of a corresponding amount of purified air at a desired temperature to each of the zones; and
 - an air handling unit (116) configured to receive the one or more command signals from the IoT-managed control device (102) regarding the quality of air in
 - 15 the one or more zones, and provide purified air to said one or more zones through air circulation lines, wherein the air handling unit (116) receives vent air (118) at a first temperature as input and provides purified air at a second temperature,
 - wherein the IoT-managed control device (102) comprises an AI/ML module (106) configured to predict air quality within the one or more zones of the
 - 20 building based on the processing of the sensor data, at least one piece of forecast data received from one or more sources over a network (122), and historical data, and optimizes the energy consumption within the building (114) based on predicted data.
2. The system (100, 200) of claim 1, wherein the air quality sensor arrays (120)
- 25 comprise at least one sensor selected from each of the groups consisting of particulate matter sensors, volatile organic compound (VOC) sensors, carbon dioxide (CO₂) sensors, nitrogen dioxide (NO₂) sensors, Sox, Ozone, Radon sensors temperature sensors, humidity sensors, and air pressure sensors.

3. The system (100, 200) of claim 1, wherein the AI/ML module (106) is configured to adaptively adjust the one or more command signals for each zone based on historical air quality data and standard preferences predefined for each of the zones.
- 5 4. The system (100, 200) of claim 1, wherein the air handling unit (116) is configured to vary the air purification level based on the severity of detected air pollutants in a zone, as indicated by the received one or more command signals for the zone.
- 10 5. The system (100, 200) of claim 1, wherein the AI/ML module (106) is configured to predict potential faults or inefficiencies in the air handling unit (116) based on the processed sensor data, facilitating preemptive maintenance to sustain optimal air purification performance.
- 15 6. The system (100, 200) of claim 1, wherein the IoT-managed control device (102) includes a user interface enabling users to monitor real-time air quality data, adjust purification settings, and receive notifications or recommendations based on the predictions made by the AI/ML module.
- 20 7. The system (100, 200) as claimed in claim 1, the system further comprises a fresh air handling unit (206) adapted to operate in both isolated and standby modes with the air handling unit (116), allowing occasional intake of fresh air into the system.
8. The system (100, 200) as claimed in claim 1, wherein the air handling unit (116) comprises a fan-coil unit (204), which in one mode, is adapted to receive stale air from the one or more zones of the building (114), at different temperatures, and condition the air using hot or cold water circulation, and in
25 another mode, purify and condition the air based on the sensor data.
9. The system (100, 200) as claimed in claim 8, wherein the air handling unit (116) utilizes either the waste heat produced within the building for heating the water or cool vent air for maintaining a temperature of water.

10. The system (100, 200) as claimed in claim 8, wherein the fan-coil unit (204) is further coupled with at least one air recycle compressor (314) adapted to receive the vent air, a knockout drum (308) adapted to separate moisture from air, and air pressure reducing valve (302) to release purified and processed air at a lower a
5 temperature to the one or more zones of the building (114) and a turbine (304) configured to receive pressurized air from the KO Drum (308) and uses the same to generate electricity.
11. The system (100, 200) as claimed in claim 1, wherein the air quality sensor arrays (120) include at least one of an exhaust air quality meter (208), a
10 continuous or clean air quality meter (210), and an indoor air quality meter (216).
12. The system (100, 200) as claimed in claim 1, wherein the system is further coupled to an energy meter (218) and a thermal unit meter (212), wherein the processing unit is configured to regulate power consumption in the building (114) based on the energy consumption data.
- 15 13. The system (100, 200) as claimed in claim 1, wherein the air quality sensor arrays (120) are installed on both an outside and an inside of the building (114), wherein the IoT-managed control device (102) is configured to trigger an alarm module in case of a variation in readings of an array in one zone compared to corresponding arrays in other zones beyond a threshold.
- 20 14. The system (100, 200) as claimed in claim 1, wherein the purified air is provided to said one or more zones through air circulation lines.
15. A method for air purification in a building using a system comprising a plurality of air quality sensor arrays (120), an IoT-managed control device (102), an air handling unit (116), the method comprising:
- 25 receiving air quality data from the plurality of air quality sensor arrays (120) installed in multiple zones within the building (114), both inside and outside of the building (114);
- processing the received air quality data using an IoT-managed control device (102) to generate one or more command signals based on the processed air

quality data from each zone, directing the supply of a corresponding amount of purified air to each of the zones;

transmitting the generated one or more command signals from the IoT-managed control device (102) to an air handling unit (116), the air handling unit
5 (116) is configured to receive the command signals regarding the quality of air in the multiple zones and provide purified air to said multiple zones through air circulation lines, wherein the air handling unit (116) receives vent air at a first temperature as input and provides purified air at a second temperature;

wherein an AI/ML module (106) to predict air quality within the multiple
10 zones of the building (114) by processing sensor data, weather forecast data, and historical weather data;

optimizing energy consumption within the building (114) based on the predictions made by the AI/ML module (106), thereby improving air quality within the multiple zones.

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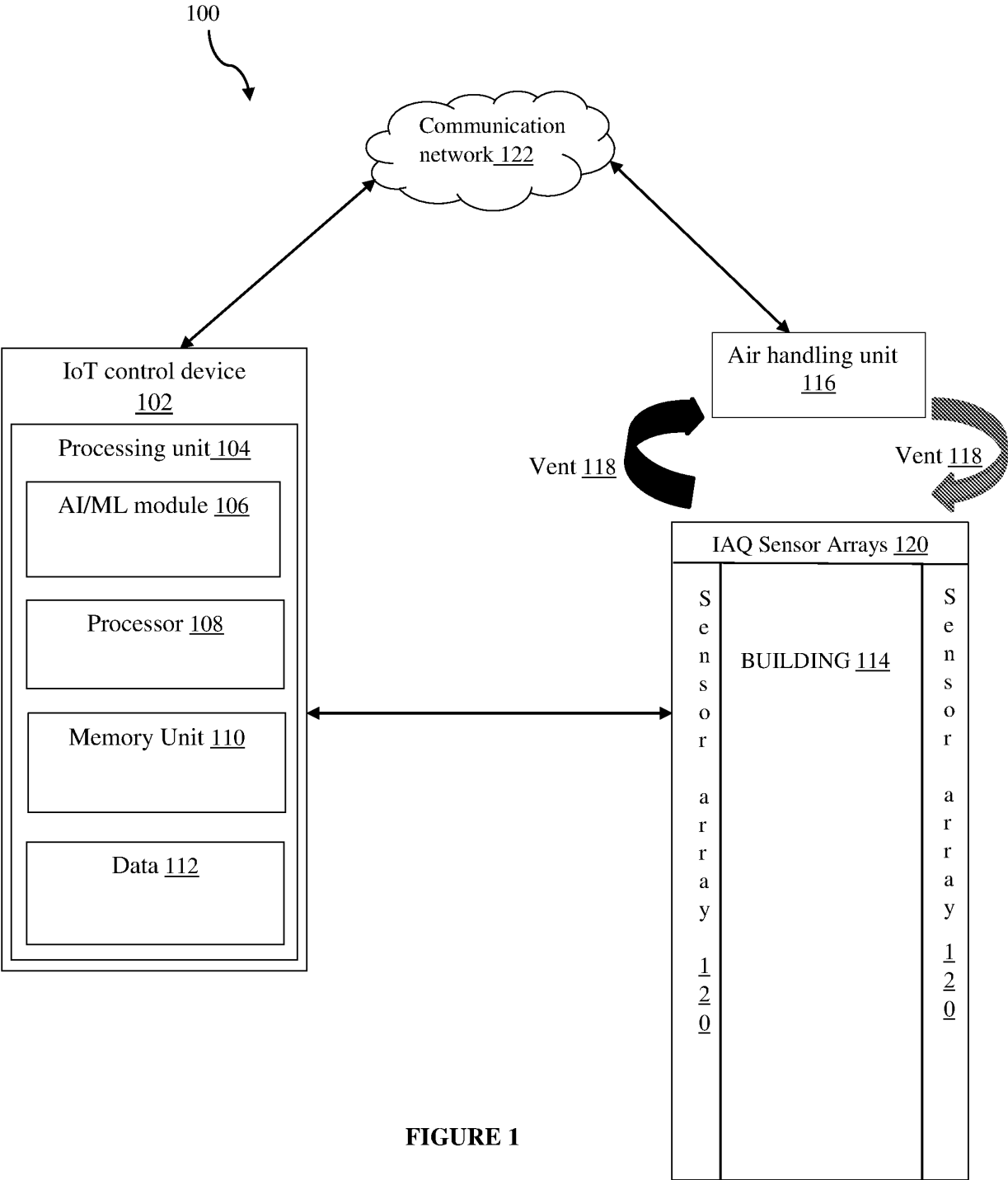


FIGURE 1

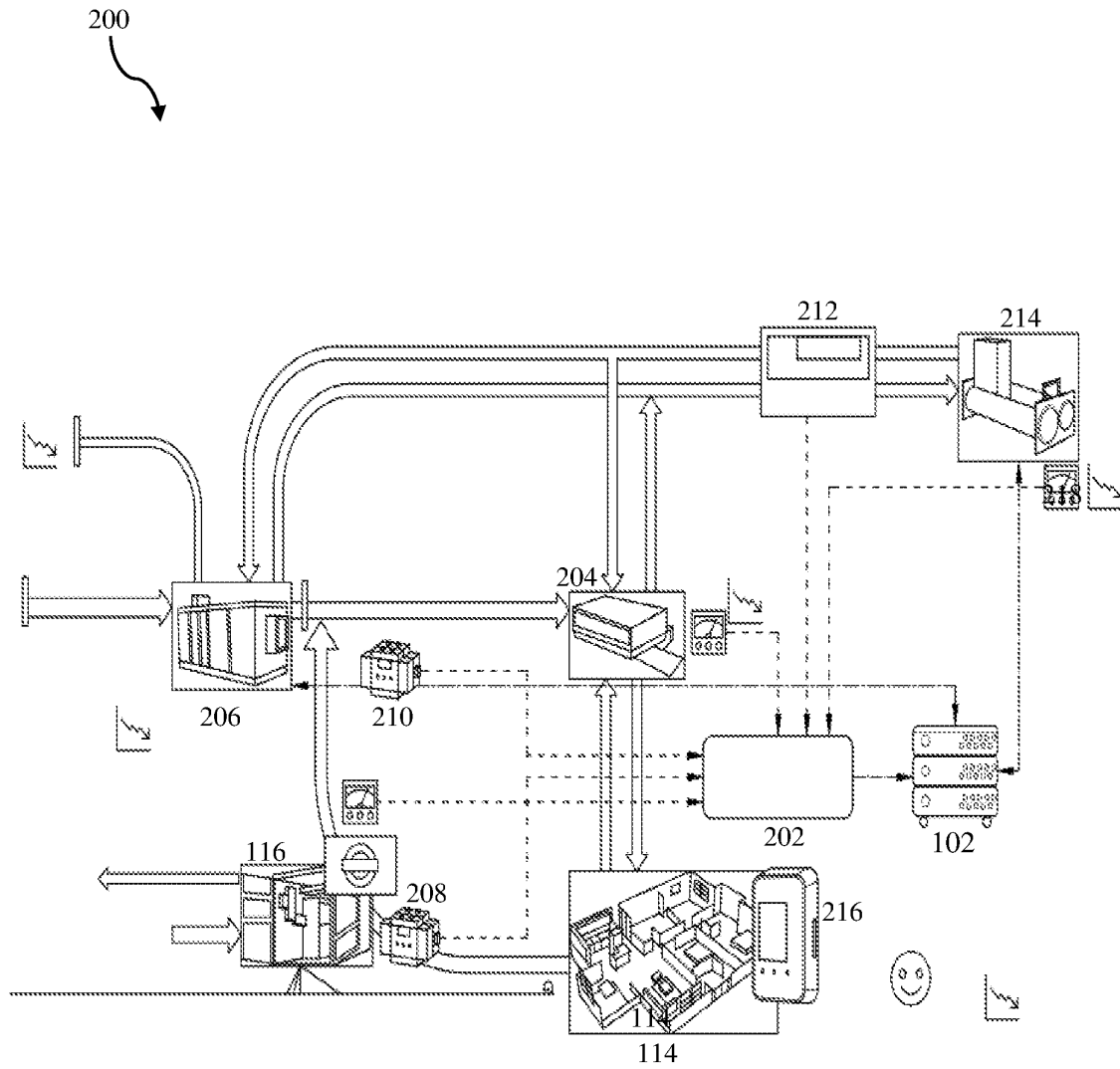


FIGURE 2

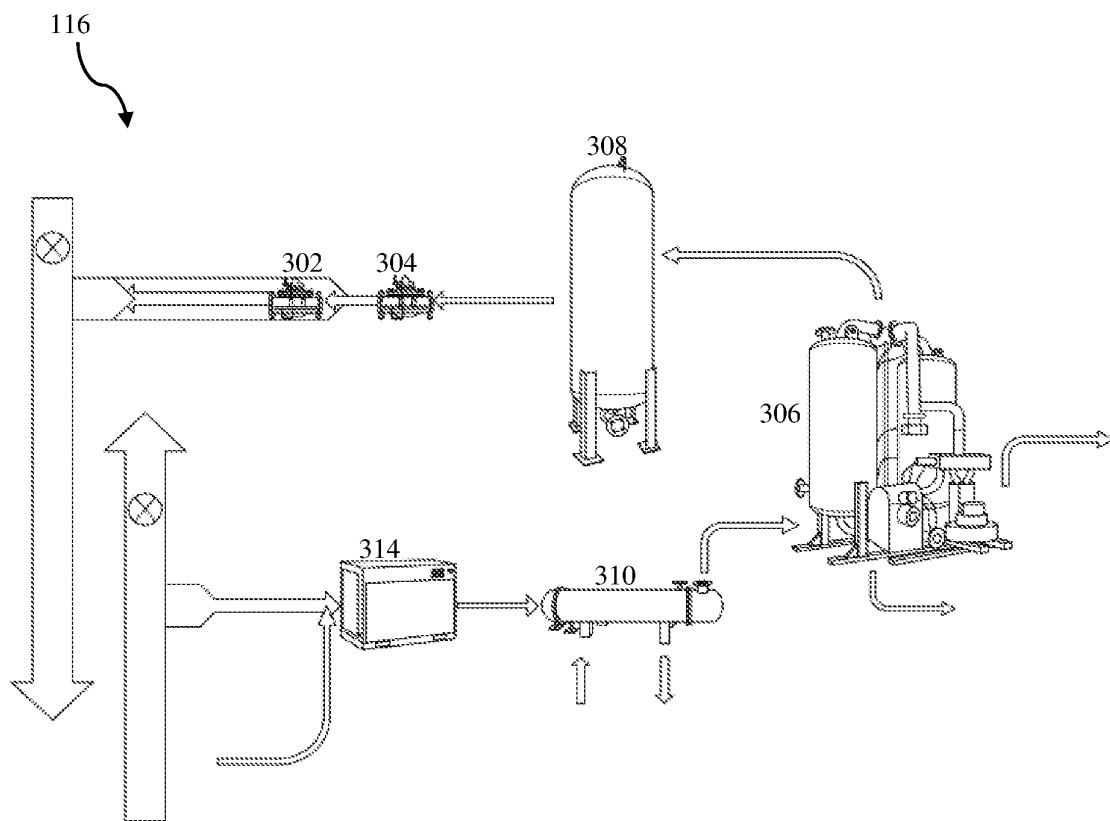


FIGURE 3

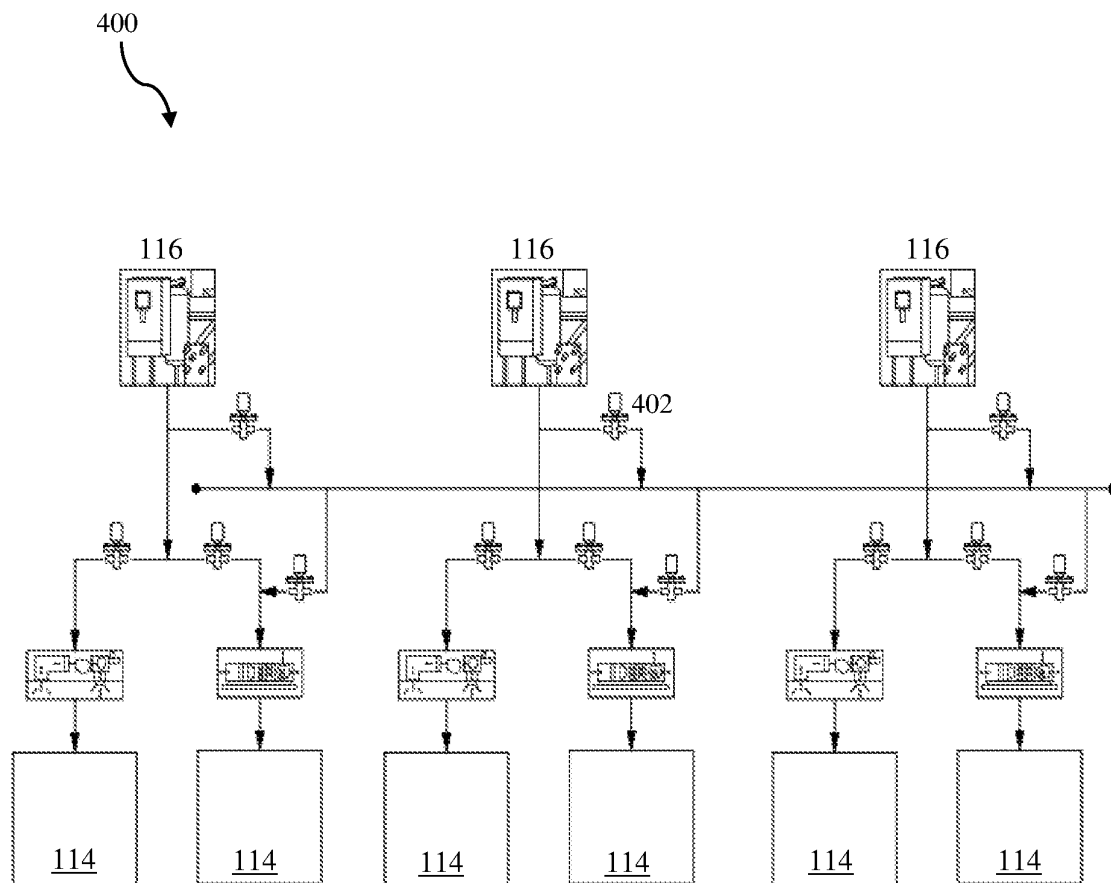


FIGURE 4

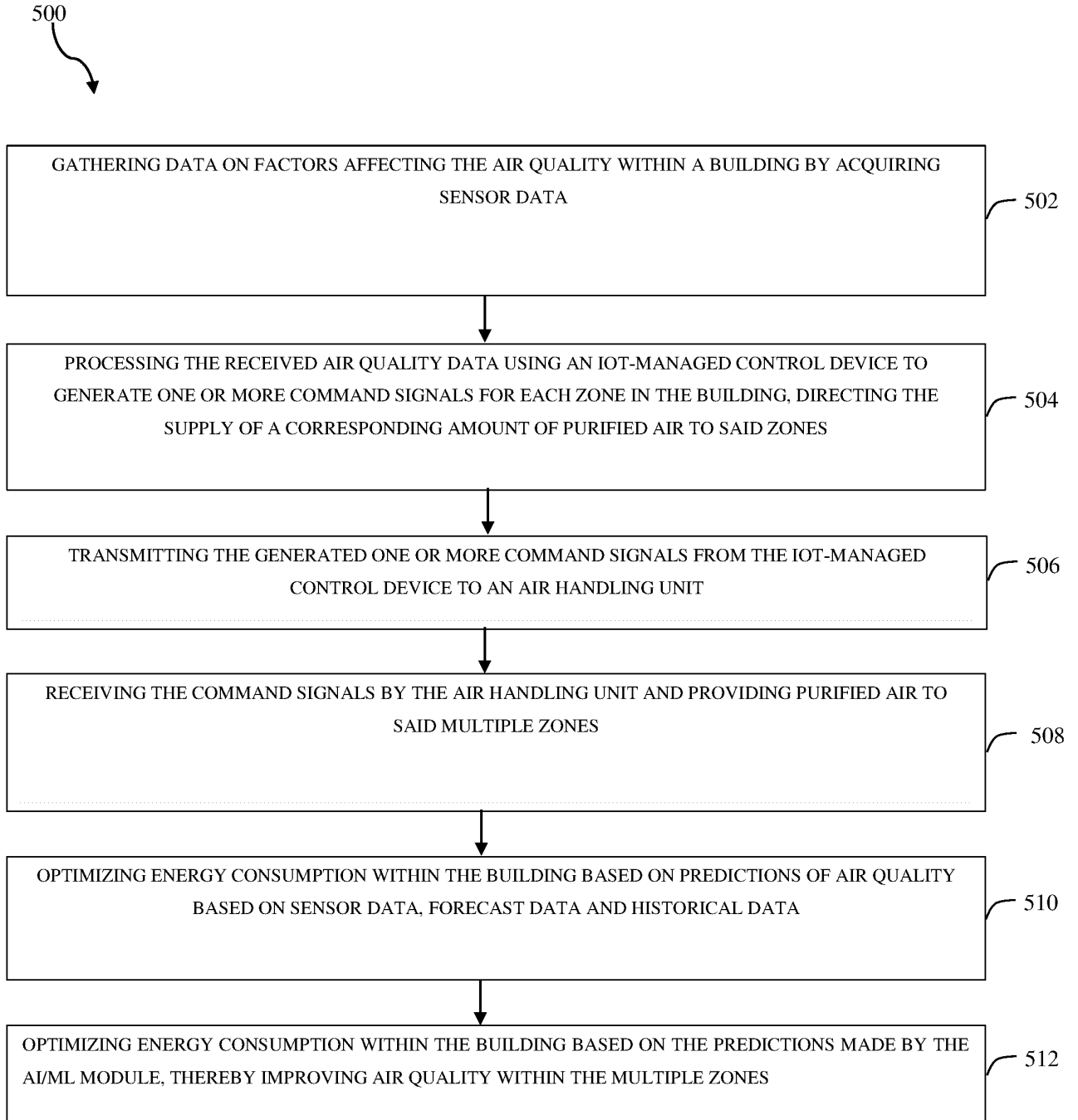


FIGURE 5

INTERNATIONAL SEARCH REPORT

International application No.

PCT/IN2024/050179

A. CLASSIFICATION OF SUBJECT MATTER

G05B13/02, G16Y20/10, G16Y40/20, F24F11/32, F24F11/63 Version=2024.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G05B, G16Y, F24F

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic database consulted during the international search (name of database and, where practicable, search terms used)

PatSeer, IPO Internal Database, Google Patents, Keywords - generate, change, synthesize, command, purified air or air, IOT, manage, control

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	US20230408477A1 (AIR RATING INC) 21 DEC 2023 (21-12-2023) Para(s). [0149-0156], [0201-0207], [0352-0358], figs. 1, 14, 15, 16, 17	1-15
Y	US20210364184A1 (PRAKASH SAVAKKANAVAR) 25 NOV 2021 (25-11-2021) Para(s). [0049-0056], [0056-0062]	1-15

☐ Further documents are listed in the continuation of Box C.
☒ See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

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"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

14-05-2024

Date of mailing of the international search report

14-05-2024

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INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/IN2024/050179

Citation	Pub.Date	Family	Pub.Date
US 20210364184 A1	25-11-2021	US 11604004 B2	14-03-2023
		WO 2021130759 A1	01-07-2021